

FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform

5. Project Development Phase

Frontend Development

Developed using React.js and Vite. Created responsive pages including Login, Register, Dashboard, Financial Health, Settlement Predictor, Negotiation Email, Know Your Rights, and History. Reusable components such as Navbar, Sidebar, Stats Cards, and Loan Table were implemented.

Backend Development

Implemented with FastAPI and Python. Developed modular REST APIs for authentication, financial profile management, loan management, AI services, borrower rights, and history tracking.

Database Implementation

SQLite database was configured to store user accounts, financial profiles, loan details, and history records using SQLAlchemy models.

Authentication Module

Implemented secure registration and login using password hashing and JWT-based authentication with protected API endpoints.

Financial Management Module

Implemented borrower financial profile management, monthly surplus calculation, debt ratio calculation, and loan CRUD operations.

AI Integration

Integrated Google Gemini AI with a rule-based fallback mechanism to generate settlement recommendations and negotiation emails when AI services are unavailable.

Error Handling & Performance

Added structured exception handling, validation, API error responses, and optimized frontend-backend communication for better performance and reliability.

Version Control

Maintained the complete project using Git and GitHub with organized commits, repository structure, and documentation.